

Exact Identity Representation from Odd High-Accuracy ReLU SGD

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1 Theoretical Setup

Fix integers $n, m, T \geq 1$, a stepsize $\eta > 0$, and an accuracy parameter $\varepsilon \geq 0$. Let $\mathcal{X} = \{-1, +1\}^n$ and let $\mathcal{H} \subseteq \{-1, +1\}^{\mathcal{X}}$. Write $\Delta(\mathcal{X})$ for the probability distributions on the finite set $\mathcal{X}$, and fix a tie label $s_0 \in \{-1, +1\}$. Throughout, scores are converted to labels by

$$\text{sign}_{s_0}(z) = \begin{cases} +1, & z > 0, \\ -1, & z < 0, \\ s_0, & z = 0. \end{cases}$$

For a score $g : \mathcal{X} \rightarrow \mathbb{R}$, distribution $\mathcal{D}$, and target $h \in \mathcal{H}$, define the strict source error

$$\mathcal{L}_{\mathcal{D},h}(g) := \Pr_{x \sim \mathcal{D}} [\text{sign}_{s_0}(g(x))h(x) < 0].$$

Thus a zero score is an error precisely at points whose target label is $-s_0$.

The learner is the bias-free depth-two fully connected ReLU network

$$f_{a,W}(x) = a^\top \sigma(Wx), \quad W \in \mathbb{R}^{m \times n}, \quad a \in \mathbb{R}^m, \quad \sigma(r) = \max\{0, r\}$$

with coordinatewise ReLU. Both layers are trainable and the parameter count is

$$S = mn + m = m(n + 1).$$

At initialization the entries are independent with

$$W_{ji}^{(0)} \sim \mathcal{N}(0, 1/n), \quad a_j^{(0)} \sim \mathcal{N}(0, 1/m).$$

For a fixed pair $(\mathcal{D}, h)$, fresh samples $x^{(t)} \sim \mathcal{D}$ are drawn for $t = 0, \dots, T - 1$ and the all-layer one-sample SGD recursion is

$$(a^{(t+1)}, W^{(t+1)}) = (a^{(t)}, W^{(t)}) - \eta \nabla_{(a,W)} \ell(h(x^{(t)})f_{a^{(t)}, W^{(t)}}(x^{(t)})), \quad \ell(z) = \log(1 + e^{-z}),$$

using the source gradient convention at ReLU kinks. For a realized initialization and sample path ω , define

$$G_\omega(x) = \sum_{t=\lceil T/2 \rceil}^T f_{a^{(t)}, W^{(t)}}(x), \quad \hat{h}_\omega(x) = \text{sign}_{s_0}(G_\omega(x)).$$

Let $\mathbb{Q}_{\mathcal{D},h}$ denote the joint law of the Gaussian initialization and the fresh SGD samples. Also set

$$\mathcal{D}^-(x) = \mathcal{D}(-x), \quad \mathcal{D}^{\text{sym}} = \frac{1}{2}(\mathcal{D} + \mathcal{D}^-),$$

and, for each realized trajectory,

$$A_\omega(x) = G_\omega(x) - G_\omega(-x), \quad v_\omega = \sum_{t=\lceil T/2 \rceil}^T (W^{(t)})^\top a^{(t)}.$$

For a feature map $\varphi : \mathcal{X} \rightarrow \mathbb{R}^d$, use the same tie-resolved homogeneous classifier $x \mapsto \text{sign}_{s_0}(\langle w, \varphi(x) \rangle)$. The deterministic dimension complexity is the least d for which one map φ satisfies

$$\forall h \in \mathcal{H} \exists w_h \in \mathbb{R}^d \forall x \in \mathcal{X}, \quad \text{sign}_{s_0}(\langle w_h, \varphi(x) \rangle) = h(x).$$

The confident dimension $\text{dc}^{1/2}(\mathcal{H})$ is the least d for which a law $\mathcal{P}$ over such maps, chosen before the later choices of $\mathcal{D}$ and h , satisfies

$$\forall \mathcal{D} \in \Delta(\mathcal{X}) \forall h \in \mathcal{H}, \quad \Pr_{\varphi \sim \mathcal{P}} \left[\inf_{w \in \mathbb{R}^d} \Pr_{x \sim \mathcal{D}} [\text{sign}_{s_0}(\langle w, \varphi(x) \rangle) h(x) < 0] = 0 \right] \geq \frac{1}{2}.$$

Finally, let $\varphi_{\text{id}} : \mathcal{X} \rightarrow \mathbb{R}^n$ be the identity map $\varphi_{\text{id}}(x) = x$.

Assumption 1 (Antipodally odd target class). *For every $h \in \mathcal{H}$ and every $x \in \mathcal{X}$,*

$$h(-x) = -h(x).$$

Assumption 2 (Strict high-accuracy regime). *The primitive accuracy condition*

$$2\varepsilon < \frac{1}{n+1}$$

holds.

Assumption 3 (Universal source success). *The architecture, width, stepsize, and horizon above are fixed before the later choices of distribution and target, and*

$$\forall \mathcal{D} \in \Delta(\mathcal{X}) \forall h \in \mathcal{H}, \quad \mathbb{E}_{\omega \sim \mathbb{Q}_{\mathcal{D},h}} [\mathcal{L}_{\mathcal{D},h}(G_\omega)] \leq \varepsilon.$$

The expectation is jointly over the independent Gaussian initialization and all fresh one-sample SGD draws; no learner parameter may depend on the later choice of $\mathcal{D}$ or h .

1.1 Formalized Goal

Under Assumptions 1, 2, and 3, prove

$$\forall h \in \mathcal{H} \exists w_h \in \mathbb{R}^n \forall x \in \mathcal{X}, \quad \text{sign}_{s_0}(\langle w_h, x \rangle) = h(x).$$

The consequences are the point-mass feature law $\delta_{\varphi_{\text{id}}}$ with exact-representation probability one and the dimension chain

$$\text{dc}^{1/2}(\mathcal{H}) \leq \text{dc}(\mathcal{H}) \leq n \leq S \leq TS.$$

2 Preliminaries

The cube involution $x \mapsto -x$ is a bijection of $\mathcal{X}$. Hence the reflected mass function $\mathcal{D}^-$ and the symmetrization $\mathcal{D}^{\text{sym}}$ introduced in Section 1 are the natural distributions for pairing antipodal inputs. The proof below uses the identity-coordinate score $x \mapsto \langle v, x \rangle$ and the same fixed tie-resolved map sign_{s_0} as the source learner. All representative, convex-hull, support, and trajectory abbreviations are introduced locally in the corresponding appendix proofs.

3 Main Theorem

Theorem 3.1 (Exact identity representation in the odd high-accuracy regime). *Under Assumptions 1, 2, and 3, the identity feature map exactly represents the entire class:*

$$\forall h \in \mathcal{H} \exists w_h \in \mathbb{R}^n \forall x \in \mathcal{X}, \quad \text{sign}_{s_0}(\langle w_h, x \rangle) = h(x).$$

Moreover, the law $\mathcal{P}_{\text{id}} := \delta_{\varphi_{\text{id}}}$ is fixed before every distribution and target and satisfies

$$\forall \mathcal{D} \in \Delta(\mathcal{X}) \forall h \in \mathcal{H}, \quad \Pr_{\varphi \sim \mathcal{P}_{\text{id}}} \left[\inf_{w \in \mathbb{R}^n} \Pr_{x \sim \mathcal{D}} [\text{sign}_{s_0}(\langle w, \varphi(x) \rangle) h(x) < 0] = 0 \right] = 1.$$

Consequently,

$$\text{dc}^{1/2}(\mathcal{H}) \leq \text{dc}(\mathcal{H}) \leq n \leq S \leq TS.$$

These conclusions also hold when $\mathcal{H} = \emptyset$, with the target quantifiers interpreted vacuously. The result is deterministic after the source premise, nonasymptotic, and uses the fixed finite horizon T . The displayed constants 1 and 2 have no hidden dependence on $n, m, S, T, \eta, \varepsilon, \mathcal{H}, \mathcal{D}, h$, initialization, or the sample path.

The theorem is a depth-two, bias-free, antipodally odd, strict-accuracy specialization. It does not assert the corresponding unrestricted-depth, non-odd, or unrestricted-accuracy statement.

4 Proof Sketch

The scalar identity $\sigma(r) - \sigma(-r) = r$ makes the antisymmetric part of each realized depth-two network exactly linear. Summing over the prescribed latter-half iterates gives $A_\omega(x) = \langle v_\omega, x \rangle$; see Proposition A.1.

Oddness and the fixed tie rule imply a pointwise pair-error comparison. If G_ω correctly labels both x and $-x$, one of the two correctness conditions is strict even when the other score is zero, so $G_\omega(x) - G_\omega(-x)$ has the strict sign of $h(x)$. Integrating this comparison gives Proposition A.2,

$$\mathcal{L}_{\mathcal{D},h}(A_\omega) \leq 2\mathcal{L}_{\mathcal{D}^{\text{sym}},h}(G_\omega).$$

The universal source premise is then invoked on the legal distribution $\mathcal{D}^{\text{sym}}$. A finite-range expectation argument extracts, for every original pair $(\mathcal{D}, h)$, a deterministic v satisfying

$$\mathcal{L}_{\mathcal{D},h}(x \mapsto \langle v, x \rangle) \leq 2\varepsilon.$$

For a fixed target, Q_h contains exactly the representative labeled $-s_0$ from each antipodal pair. Exact homogeneous representation is equivalent to feasibility of the strict system $h(q)\langle w, q \rangle > 0$ on Q_h . If this system were infeasible, the closest point of the convex hull of the signed vectors $h(q)q$ would be zero. Affine-dependence pruning reduces the resulting positive convex certificate to at most $n + 1$ representatives. On the uniform distribution over those representatives, every homogeneous score makes at least one tie-resolved error, so its risk is at least $1/(n + 1)$.

Applying the deterministic 2ε upper bound to that same witness distribution contradicts Assumption 2. Thus the strict system is feasible for every target and the identity map represents the class exactly. The Dirac law on the identity map then succeeds with probability one. Finally, every deterministic representation induces a confident one by taking a point mass, and $n \leq m(n + 1) = S \leq TS$, which yields the asserted chain.

A Proof Details

A.1 Exact ReLU Antisymmetrization

Lemma A.1 (Per-iterate ReLU antisymmetrization). *For the bias-free depth-two architecture of Section 1, every $W \in \mathbb{R}^{m \times n}$, $a \in \mathbb{R}^m$, and $x \in \mathcal{X}$ satisfy*

$$f_{a,W}(x) - f_{a,W}(-x) = \langle W^\top a, x \rangle.$$

No regularity, boundedness, or nonzero condition on a or W is required.

Proof. Let $r_j = (Wx)_j$ for $j = 1, \dots, m$. Linearity gives $W(-x) = -Wx$, and hence

$$f_{a,W}(x) - f_{a,W}(-x) = \sum_{j=1}^m a_j (\sigma(r_j) - \sigma(-r_j)).$$

For every real r ,

$$\sigma(r) - \sigma(-r) = \max\{0, r\} - \max\{0, -r\} = r.$$

If $r > 0$, the two ReLU values are r and 0; if $r < 0$, they are 0 and $-r$; and if $r = 0$, both are zero. Therefore

$$\sum_{j=1}^m a_j (\sigma(r_j) - \sigma(-r_j)) = \sum_{j=1}^m a_j r_j = a^\top Wx = \langle W^\top a, x \rangle.$$

The calculation remains valid when any $r_j = 0$, when $a = 0$, or when $W = 0$; it uses no derivative or subgradient at a ReLU kink. $\square$

Proposition A.1 (Exact latter-half aggregate linearization). *For the network and aggregate defined in Section 1, every realized trajectory ω and every $x \in \mathcal{X}$ satisfy*

$$A_\omega(x) = G_\omega(x) - G_\omega(-x) = \left\langle \sum_{t=\lceil T/2 \rceil}^T (W^{(t)})^\top a^{(t)}, x \right\rangle = \langle v_\omega, x \rangle.$$

Proof. Let

$$I_T := \{t \in \mathbb{Z} : \lceil T/2 \rceil \leq t \leq T\}.$$

Because $T \geq 1$, this is a finite nonempty integer interval. Subtracting the two aggregate sums term by term gives

$$\begin{aligned} A_\omega(x) &= \sum_{t \in I_T} f_{a^{(t)}, W^{(t)}}(x) - \sum_{t \in I_T} f_{a^{(t)}, W^{(t)}}(-x) \\ &= \sum_{t \in I_T} (f_{a^{(t)}, W^{(t)}}(x) - f_{a^{(t)}, W^{(t)}}(-x)). \end{aligned}$$

Applying Lemma A.1 separately to every listed state and using finite linearity of the inner product yields

$$\begin{aligned} A_\omega(x) &= \sum_{t \in I_T} \langle (W^{(t)})^\top a^{(t)}, x \rangle \\ &= \left\langle \sum_{t \in I_T} (W^{(t)})^\top a^{(t)}, x \right\rangle = \langle v_\omega, x \rangle. \end{aligned}$$

Each state is merely an arbitrary finite pair $(a^{(t)}, W^{(t)})$, so the argument is valid at initialization, after any update, at every kink, and for zero parameter states. It invokes neither the SGD recursion nor its gradient convention. When $T = 1$, $I_T = \{1\}$, exactly the single returned latter-half iterate is present. Thus the equality is pathwise and has zero residual for every realization. $\square$

Consequently, for every realized trajectory and every cube point, the antisymmetric aggregate is the identity-coordinate score $x \mapsto \langle v_\omega, x \rangle$. This exact equality is the sole score interface used in the following error-transfer argument; it remains valid in the zero-parameter and exact/noiseless regimes.

A.2 Tie-Aware Antipodal Error Transfer

For a score $F : \mathcal{X} \rightarrow \mathbb{R}$, define locally

$$e_{F,h}(z) := \mathbf{1}\{\text{sign}_{s_0}(F(z))h(z) < 0\}.$$

Since both labels are in $\{-1, +1\}$, this indicator is zero exactly when $\text{sign}_{s_0}(F(z)) = h(z)$.

Lemma A.2 (Tie-aware antipodal pair-error comparison). *Under Assumption 1, for every $h \in \mathcal{H}$, realized trajectory ω , and $x \in \mathcal{X}$,*

$$e_{A_\omega,h}(x) \leq e_{G_\omega,h}(x) + e_{G_\omega,h}(-x), \quad A_\omega(x) = G_\omega(x) - G_\omega(-x).$$

Proof. Fix the stated objects and abbreviate

$$y = h(x), \quad u = G_\omega(x), \quad q = G_\omega(-x).$$

Assumption 1 gives $h(-x) = -y$. Suppose first that both source indicators on the right vanish. Then u is correctly classified with label y , and q is correctly classified with label $-y$. The four possible pairs (s_0, y) give

s_0	y	condition on u	condition on q	consequence for $u - q$
+1	+1	$u \geq 0$	$q < 0$	$u - q > 0$
+1	-1	$u < 0$	$q \geq 0$	$u - q < 0$
-1	+1	$u > 0$	$q \leq 0$	$u - q > 0$
-1	-1	$u \leq 0$	$q > 0$	$u - q < 0$.

Thus $y(u - q) > 0$ in every row. One correctness condition may be weak because its target is the tie label, but the correctness condition for the opposite label is strict. Consequently $A_\omega(x) = u - q$ is nonzero and has label y , so $e_{A_\omega,h}(x) = 0$.

This table also exhausts all source-score ties. If $u = 0$ is correct, then $y = s_0$ and correctness of q for $-y$ is strict. If $q = 0$ is correct, then $-y = s_0$ and correctness of u for $y = -s_0$ is strict. A zero at a point labeled $-s_0$ is already a source error. If $u = q = 0$, exactly one of the opposite labels y and $-y$ equals s_0 , so exactly one source indicator is one.

It remains to check a tie of the antisymmetric score. If $A_\omega(x) = 0$ and $y = s_0$, its indicator is zero. If $A_\omega(x) = 0$ and $y = -s_0$, its indicator is one, while simultaneous source correctness is impossible because the preceding calculation would force $yA_\omega(x) > 0$. Hence at least one source indicator is one. In every remaining case where a source indicator is one, the right side is at least one and the left side, being an indicator, is at most one. This proves the comparison in every zero and nonzero branch. $\square$

Proposition A.2 (Exact factor-two risk transfer). *Under Assumption 1 and Lemma A.2, every $\mathcal{D} \in \Delta(\mathcal{X})$, $h \in \mathcal{H}$, and realized trajectory ω satisfy*

$$\mathcal{L}_{\mathcal{D},h}(A_\omega) \leq \mathcal{L}_{\mathcal{D},h}(G_\omega) + \mathcal{L}_{\mathcal{D}^-,h}(G_\omega) = 2\mathcal{L}_{\mathcal{D}^{\text{sym}},h}(G_\omega).$$

Proof. Multiply the pointwise inequality in Lemma A.2 by $\mathcal{D}(x)$ and sum over the finite cube:

$$\begin{aligned}\mathcal{L}_{\mathcal{D},h}(A_\omega) &= \sum_{x \in \mathcal{X}} \mathcal{D}(x) e_{A_\omega,h}(x) \\ &\leq \sum_{x \in \mathcal{X}} \mathcal{D}(x) e_{G_\omega,h}(x) + \sum_{x \in \mathcal{X}} \mathcal{D}(x) e_{G_\omega,h}(-x) \\ &= \mathcal{L}_{\mathcal{D},h}(G_\omega) + \sum_{x \in \mathcal{X}} \mathcal{D}(x) e_{G_\omega,h}(-x).\end{aligned}$$

The last indicator evaluates both the score and target at $-x$. Since $x \mapsto -x$ is a bijection, the substitution $z = -x$ gives

$$\begin{aligned}\sum_{x \in \mathcal{X}} \mathcal{D}(x) e_{G_\omega,h}(-x) &= \sum_{z \in \mathcal{X}} \mathcal{D}(-z) e_{G_\omega,h}(z) \\ &= \sum_{z \in \mathcal{X}} \mathcal{D}^-(z) e_{G_\omega,h}(z) = \mathcal{L}_{\mathcal{D}^-,h}(G_\omega).\end{aligned}$$

Finally,

$$\begin{aligned}2\mathcal{L}_{\mathcal{D}^{\text{sym}},h}(G_\omega) &= 2 \sum_{z \in \mathcal{X}} \frac{\mathcal{D}(z) + \mathcal{D}^-(z)}{2} e_{G_\omega,h}(z) \\ &= \mathcal{L}_{\mathcal{D},h}(G_\omega) + \mathcal{L}_{\mathcal{D}^-,h}(G_\omega).\end{aligned}$$

No symmetry of $\mathcal{D}$ is assumed. The pointwise zero-score checks in Lemma A.2 remain intact under the finite sum, and the numerical factor is exactly 2. $\square$

Together, Propositions A.1 and A.2 attach the last risk inequality to the exact homogeneous score $x \mapsto \langle v_\omega, x \rangle$, pathwise for every realization and with no trajectory condition.

In particular, the pointwise comparison holds for both tie labels, all four target/tie combinations, source-score zeros, and $A_\omega(x) = 0$. Summing under any original distribution and changing variables by the cube involution therefore gives exactly the factor-two symmetrized risk, with no symmetry assumption on the original distribution.

A.3 Distributional Extraction of an Approximate Separator

Lemma A.3 (Legality of antipodal symmetrization). *For every $\mathcal{D} \in \Delta(\mathcal{X})$, $\mathcal{D}^{\text{sym}} = (\mathcal{D} + \mathcal{D}^-)/2$ belongs to $\Delta(\mathcal{X})$.*

Proof. Every mass in $\mathcal{D}^{\text{sym}}$ is nonnegative. The cube involution is a bijection, so

$$\sum_{x \in \mathcal{X}} \mathcal{D}^-(x) = \sum_{x \in \mathcal{X}} \mathcal{D}(-x) = \sum_{z \in \mathcal{X}} \mathcal{D}(z) = 1.$$

Consequently,

$$\sum_{x \in \mathcal{X}} \mathcal{D}^{\text{sym}}(x) = \frac{1}{2} \left(\sum_x \mathcal{D}(x) + \sum_x \mathcal{D}^-(x) \right) = 1,$$

which is exactly the probability-distribution condition. $\square$

Proposition A.3 (Expected factor-two identity-coordinate transfer). *Under Assumptions 1 and 3, for every $\mathcal{D} \in \Delta(\mathcal{X})$ and $h \in \mathcal{H}$,*

$$\mathbb{E}_{\omega \sim \mathbb{Q}_{\mathcal{D}^{\text{sym}},h}} [\mathcal{L}_{\mathcal{D},h}(x \mapsto \langle v_\omega, x \rangle)] \leq 2\varepsilon.$$

Proof. Fix $\mathcal{D}$ and h . Lemma A.3 makes $\mathcal{D}^{\text{sym}}$ a legal input distribution. For every trajectory under $\mathbb{Q}_{\mathcal{D}^{\text{sym}}, h}$, Proposition A.1 gives

$$A_\omega(x) = \langle v_\omega, x \rangle \quad (x \in \mathcal{X}),$$

so the corresponding losses under the original evaluation law are equal. Applying Proposition A.2 with evaluation law $\mathcal{D}$ gives the corresponding pathwise inequality. The Gaussian initialization coordinates and the finitely many sampled cube points are measurable random variables. Under the fixed source gradient convention, each one-step SGD update is a Borel function of the current coordinates and the current sample; induction over the finite horizon therefore makes every coordinate of $(a^{(t)}, W^{(t)})$ measurable. Consequently v_ω is a finite sum of algebraic functions of measurable coordinates. For each fixed x , the score $\langle v_\omega, x \rangle$ is measurable, the fixed tie-resolved error indicator is Borel, and

$$\mathcal{L}_{\mathcal{D}, h}(x \mapsto \langle v_\omega, x \rangle) = \sum_{x \in \mathcal{X}} \mathcal{D}(x) \mathbf{1}\{\text{sign}_{s_0}(\langle v_\omega, x \rangle)h(x) < 0\}$$

is a finite sum of Borel random variables. The same finite-state argument makes the source loss measurable. Both losses are bounded, so taking expectation under this same trajectory law gives

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}_{\mathcal{D}^{\text{sym}}, h}} \mathcal{L}_{\mathcal{D}, h}(x \mapsto \langle v_\omega, x \rangle) &\leq 2\mathbb{E}_{\mathbb{Q}_{\mathcal{D}^{\text{sym}}, h}} \mathcal{L}_{\mathcal{D}^{\text{sym}}, h}(G_\omega) \\ &\leq 2\varepsilon. \end{aligned}$$

The second inequality is Assumption 3 instantiated at $(\mathcal{D}^{\text{sym}}, h)$. Both losses are bounded in $[0, 1]$ and the displayed pathwise inequality uses exactly the same law on both sides; no expectation or distribution is interchanged. $\square$

Lemma A.4 (Finite-domain expectation-to-existence). *Fix $\mathcal{D} \in \Delta(\mathcal{X})$ and $h \in \mathcal{H}$. Let $\mathbb{Q}$ be a probability law for which v_ω and $\mathcal{L}_{\mathcal{D}, h}(x \mapsto \langle v_\omega, x \rangle)$ are measurable. If $c \geq 0$ and*

$$\mathbb{E}_{\omega \sim \mathbb{Q}} [\mathcal{L}_{\mathcal{D}, h}(x \mapsto \langle v_\omega, x \rangle)] \leq c,$$

then there is a deterministic $v \in \mathbb{R}^n$ with

$$\mathcal{L}_{\mathcal{D}, h}(x \mapsto \langle v, x \rangle) \leq c.$$

Proof. Define the finite subset-sum set

$$\mathcal{R}_{\mathcal{D}} := \left\{ \sum_{x \in \mathcal{X}} \mathcal{D}(x) b_x : (b_x)_{x \in \mathcal{X}} \in \{0, 1\}^{\mathcal{X}} \right\}.$$

It has at most $2^{|\mathcal{X}|}$ elements. The random variable

$$Y(\omega) := \mathcal{L}_{\mathcal{D}, h}(x \mapsto \langle v_\omega, x \rangle)$$

is measurable by hypothesis and takes values in $\mathcal{R}_{\mathcal{D}}$, because each finite-domain error indicator is either zero or one, including at a score tie. The events $\{Y = r\}$ partition the trajectory space, so

$$\mathbb{E}_{\mathbb{Q}} Y = \sum_{r \in \mathcal{R}_{\mathcal{D}}} r \mathbb{Q}[Y = r], \quad \sum_{r \in \mathcal{R}_{\mathcal{D}}} \mathbb{Q}[Y = r] = 1.$$

If no realized trajectory had $Y(\omega) \leq c$, every value with positive probability would be strictly larger than c . The finite nonempty set of such values would have a minimum $\gamma > c$, forcing $\mathbb{E}_{\mathbb{Q}} Y \geq \gamma > c$, a contradiction. Hence some value $r \leq c$ has positive probability. Its event is nonempty; choose a trajectory ω_0 in it and set $v = v_{\omega_0}$.

When $c = 0$, nonnegativity forces a positive-probability value equal to zero, so the selected vector has zero risk. When $c \geq 1$, every risk is already at most $1 \leq c$. Thus no atom, margin, or limit argument is needed. $\square$

Proposition A.4 (Distribution-wise approximate homogeneous separator). *Under Assumptions 1 and 3, for every $\mathcal{D} \in \Delta(\mathcal{X})$ and $h \in \mathcal{H}$ there exists a deterministic $v = v(\mathcal{D}, h) \in \mathbb{R}^n$ such that*

$$\mathcal{L}_{\mathcal{D}, h}(x \mapsto \langle v, x \rangle) \leq 2\varepsilon.$$

Proof. For the finite SGD recursion under the fixed source convention, the measurability argument in the proof of Proposition A.3 shows that v_ω and the displayed finite-domain tie-resolved loss are measurable under $\mathbb{Q}_{\mathcal{D}^{\text{sym}}, h}$. Apply that proposition and then Lemma A.4 with $\mathbb{Q} = \mathbb{Q}_{\mathcal{D}^{\text{sym}}, h}$ and $c = 2\varepsilon$. The resulting vector may depend on the displayed pair $(\mathcal{D}, h)$, but the identity coordinates are fixed and no common trajectory is asserted. $\square$

Thus the quantifier order is $\forall \mathcal{D} \forall h \exists v(\mathcal{D}, h)$, and the learner law used for that pair is exactly $\mathbb{Q}_{\mathcal{D}^{\text{sym}}, h}$. The architecture, stepsize, and finite horizon remain fixed. When $\varepsilon = 0$, the extracted vector has zero risk; when $\mathcal{H} = \emptyset$, the target assertion is vacuous.

A.4 Antipodal Representatives and Strict Feasibility

For each fixed target $h \in \mathcal{H}$, define the appendix-local set

$$Q_h := \{q \in \mathcal{X} : h(q) = -s_0\}.$$

The next lemma proves the representative property used by the subsequent strict-system and convex-obstruction results.

Lemma A.5 (Antipodal representative selection). *Under Assumption 1, for every fixed $h \in \mathcal{H}$ the set $Q_h = \{q : h(q) = -s_0\}$ contains exactly one member of each pair $\{x, -x\}$, and*

$$\mathcal{X} = \bigsqcup_{q \in Q_h} \{q, -q\}.$$

Proof. For $n \geq 1$, no cube point equals its antipode: $x = -x$ would force every coordinate to be zero, contrary to $x_i \in \{-1, +1\}$. Fix x and write $y = h(x)$. Oddness gives $h(-x) = -y$. Exactly one of the two opposite labels y and $-y$ equals $-s_0$, so exactly one of $x, -x$ lies in Q_h . The two-element orbits are disjoint and cover the cube, proving the partition; for $n = 1$ this is the single orbit $\{(+1), (-1)\}$. $\square$

Lemma A.6 (Pairwise strict-sign criterion). *Under Assumption 1, if $q \in Q_h$ and $w \in \mathbb{R}^n$, then*

$$[\text{sign}_{s_0}(\langle w, q \rangle) = h(q) \text{ and } \text{sign}_{s_0}(\langle w, -q \rangle) = h(-q)] \iff h(q)\langle w, q \rangle > 0.$$

Proof. Set $z = \langle w, q \rangle$. Homogeneity gives $\langle w, -q \rangle = -z$. Since $q \in Q_h$, $h(q) = -s_0$; oddness gives $h(-q) = s_0$. If $s_0 = +1$, then $h(q) = -1$ and $\text{sign}_{+1}(z) = h(q)$ iff $z < 0$, which is iff $h(q)z > 0$. If $s_0 = -1$, then $h(q) = +1$ and $\text{sign}_{-1}(z) = h(q)$ iff $z > 0$, again iff $h(q)z > 0$. Thus

$$\text{sign}_{s_0}(z) = h(q) \iff h(q)z > 0.$$

In particular $z = 0$ is an error at q . The forward implication follows immediately. Conversely, if $h(q)z > 0$, then $z \neq 0$ and

$$h(-q)\langle w, -q \rangle = (-h(q))(-z) = h(q)z > 0,$$

so both nonzero scores have their correct labels. This proves the equivalence, including both tie labels and the zero-score boundary. $\square$

Proposition A.5 (Strict-system representation equivalence). *Under Assumption 1, for every fixed $h \in \mathcal{H}$,*

$$\begin{aligned} \exists w \in \mathbb{R}^n \ \forall x \in \mathcal{X}, \quad \text{sign}_{s_0}(\langle w, x \rangle) = h(x) \\ \iff \exists w \in \mathbb{R}^n \ \forall q \in Q_h, \quad h(q)\langle w, q \rangle > 0. \end{aligned}$$

Proof. By Lemma A.5, the pairs indexed by Q_h partition $\mathcal{X}$. Exactness on all of $\mathcal{X}$ implies exactness on both members of each pair, and Lemma A.6 then gives the strict inequality on its representative. Conversely, the strict inequalities give exactness on each pair by the same lemma, and the partition extends it to every cube point. $\square$

A.5 Finite Convex Obstruction

Fix $h \in \mathcal{H}$ and write

$$Z_h := \{h(q)q : q \in Q_h\} \subseteq \mathbb{R}^n.$$

By Lemma A.5, Q_h is finite and nonempty. The strict-system equivalence in Proposition A.5 identifies infeasibility of exact representation with the absence of a vector having positive inner product with every member of Z_h .

Proposition A.6 (Finite closest-point zero certificate). *Under Assumption 1, fix $h \in \mathcal{H}$ and set*

$$Q_h := \{q \in \mathcal{X} : h(q) = -s_0\}, \quad Z_h := \{h(q)q : q \in Q_h\}.$$

Lemma A.5 makes Q_h finite and nonempty. If there is no $w \in \mathbb{R}^n$ such that $\langle w, z \rangle > 0$ for every $z \in Z_h$, then there are coefficients $\lambda_q \geq 0$ with $\sum_{q \in Q_h} \lambda_q = 1$ and

$$\sum_{q \in Q_h} \lambda_q h(q)q = 0.$$

Proof. Enumerate $Q_h = \{q_1^0, \dots, q_r^0\}$ and put $z_j = h(q_j^0)q_j^0$. Every z_j is nonzero and has norm $\sqrt{n}$. Let

$$\Delta_r = \{\lambda \in \mathbb{R}^r : \lambda_j \geq 0, \sum_{j=1}^r \lambda_j = 1\}, \quad C = \left\{ \sum_{j=1}^r \lambda_j z_j : \lambda \in \Delta_r \right\}.$$

The simplex is nonempty and compact, and the continuous function $\lambda \mapsto \|\sum_j \lambda_j z_j\|_2$ attains a minimum. Let λ^* be a minimizer and $p = \sum_j \lambda_j^* z_j$.

For fixed j and $t \in (0, 1]$, the point $p_t = (1-t)p + tz_j$ belongs to C , since its coefficients are $(1-t)\lambda_\ell^* + t\mathbf{1}\{\ell = j\}$. Minimality of p gives

$$0 \leq \|p_t\|_2^2 - \|p\|_2^2 = 2t\langle p, z_j - p \rangle + t^2\|z_j - p\|_2^2.$$

After division by t and passage to $t \downarrow 0$,

$$\langle p, z_j \rangle \geq \|p\|_2^2 \quad (1 \leq j \leq r).$$

If $p \neq 0$, then $w = p$ has $\langle w, z_j \rangle \geq \|p\|_2^2 > 0$ for every j , contradicting the local infeasibility hypothesis. Hence $p = 0$, and the minimizing coefficients give the asserted nonnegative unit-sum representation. $\square$

Lemma A.7 (Positive support pruning). *Suppose $u_1, \dots, u_R \in \mathbb{R}^n$ and $\rho_j \geq 0$ satisfy*

$$\sum_{j=1}^R \rho_j = 1, \quad \sum_{j=1}^R \rho_j u_j = 0.$$

Then there are distinct vectors $\tilde{u}_1, \dots, \tilde{u}_k$ from the original list and coefficients $\alpha_i > 0$ such that

$$1 \leq k \leq n+1, \quad \sum_{i=1}^k \alpha_i = 1, \quad \sum_{i=1}^k \alpha_i \tilde{u}_i = 0.$$

If, for a fixed $h \in \mathcal{H}$ and $Q_h := \{q \in \mathcal{X} : h(q) = -s_0\}$, each retained input satisfies $u_j = h(q_j)q_j$ for some $q_j \in Q_h$, one may retain one such q_j for each distinct $\tilde{u}_i$.

Proof. Delete all indices with $\rho_j = 0$; at least one coefficient remains. If vector values are duplicated, group equal copies and replace their coefficients by the grouped sums. The resulting distinct list $u_1, \dots, u_{k_0}$ has coefficients $\beta_i > 0$ with

$$\sum_i \beta_i = 1, \quad \sum_i \beta_i u_i = 0.$$

If $k_0 \leq n+1$, this is the desired representation. Otherwise the augmented vectors $(u_i, 1) \in \mathbb{R}^{n+1}$ are linearly dependent, so a nonzero $\gamma \in \mathbb{R}^{k_0}$ satisfies

$$\sum_i \gamma_i u_i = 0, \quad \sum_i \gamma_i = 0.$$

Because the coordinates of γ sum to zero, there are both positive and negative coordinates. Let $I_+ = \{i : \gamma_i > 0\}$ and define

$$\tau = \min_{i \in I_+} \frac{\beta_i}{\gamma_i} > 0, \quad \beta'_i = \beta_i - \tau \gamma_i.$$

For $i \in I_+$, $\beta'_i \geq 0$ and at least one is zero. For $\gamma_i < 0$, $\beta'_i > 0$, and for $\gamma_i = 0$, $\beta'_i = \beta_i > 0$. Moreover,

$$\sum_i \beta'_i = 1, \quad \sum_i \beta'_i u_i = 0.$$

Delete the newly zero coefficients and repeat. Each repetition strictly reduces the finite support, so the process terminates with at most $n+1$ positive coefficients and preserves both displayed identities. No new vector is introduced. A formal singleton support is allowed by the algebra; in the present application it cannot occur because every $h(q)q$ has norm $\sqrt{n} > 0$, but the lower-bound argument below remains valid for $k = 1$. $\square$

Lemma A.8 (Signed cancellation). *If $z_1, \dots, z_k \in \mathbb{R}^n$ and*

$$\alpha_i > 0, \quad \sum_{i=1}^k \alpha_i = 1, \quad \sum_{i=1}^k \alpha_i z_i = 0,$$

then for every $w \in \mathbb{R}^n$ there is an i with $\langle w, z_i \rangle \leq 0$.

Proof. Taking the inner product with w gives

$$0 = \left\langle w, \sum_{i=1}^k \alpha_i z_i \right\rangle = \sum_{i=1}^k \alpha_i \langle w, z_i \rangle.$$

If every inner product were strictly positive, positivity of every α_i would make the finite sum strictly positive, a contradiction. Thus one inner product is nonpositive. This includes $w = 0$ and equality cases without replacing ≤ 0 by a strict inequality. $\square$

Proposition A.7 (Uniform tie-aware obstruction). *Under Assumption 1, fix $h \in \mathcal{H}$, set $Q_h := \{q \in \mathcal{X} : h(q) = -s_0\}$, and use the representative conclusion of Lemma A.5. Suppose distinct $q_1, \dots, q_k \in Q_h$ and positive coefficients satisfy*

$$1 \leq k \leq n+1, \quad \sum_{i=1}^k \alpha_i = 1, \quad \sum_{i=1}^k \alpha_i h(q_i) q_i = 0.$$

Define

$$\mathcal{D}_h^*(x) = \begin{cases} 1/k, & x \in \{q_1, \dots, q_k\}, \\ 0, & \text{otherwise.} \end{cases}$$

Then $\mathcal{D}_h^* \in \Delta(\mathcal{X})$ and, for every $w \in \mathbb{R}^n$,

$$\mathcal{L}_{\mathcal{D}_h^*, h}(x \mapsto \langle w, x \rangle) \geq \frac{1}{k} \geq \frac{1}{n+1}.$$

Proof. Distinctness makes the displayed masses nonnegative and summing to one. Put $z_i = h(q_i)q_i$. Lemma A.8 gives an index with $\langle w, z_i \rangle \leq 0$, or equivalently

$$h(q_i) \langle w, q_i \rangle \leq 0.$$

If this product is negative, the nonzero score has prediction $-h(q_i)$ and is an error. If it is zero, then $q_i \in Q_h$ gives $h(q_i) = -s_0$, while the zero score receives label $s_0 = -h(q_i)$, again an error. This explicitly handles both choices of s_0 . Therefore at least one of the k uniform atoms is an error, and

$$\mathcal{L}_{\mathcal{D}_h^*, h}(x \mapsto \langle w, x \rangle) = \frac{1}{k} \sum_{i=1}^k \mathbf{1}\{\text{sign}_{s_0}(\langle w, q_i \rangle) h(q_i) < 0\} \geq \frac{1}{k}.$$

Since $k \leq n+1$, $1/k \geq 1/(n+1)$. For $w = 0$ every atom is a tie and the actual error is one. The same calculation covers $k = 1$, $k = n+1$, and $n = 1$. $\square$

Proposition A.8 (Finite obstruction under strict-system infeasibility). *Under Assumption 1, fix $h \in \mathcal{H}$ and set*

$$Q_h := \{q \in \mathcal{X} : h(q) = -s_0\}, \quad Z_h := \{h(q)q : q \in Q_h\}.$$

Use the representative conclusion of Lemma A.5. If there is no $w \in \mathbb{R}^n$ satisfying

$$h(q) \langle w, q \rangle > 0 \quad \text{for every } q \in Q_h$$

(equivalently, no w has positive inner product with every $z \in Z_h$), then there are distinct representatives $q_1, \dots, q_k \in Q_h$, positive coefficients with $1 \leq k \leq n+1$, and a distribution $\mathcal{D}_h^$ such that*

$$\sum_{i=1}^k \alpha_i = 1, \quad \sum_{i=1}^k \alpha_i h(q_i) q_i = 0, \quad \forall w \in \mathbb{R}^n, \quad \mathcal{L}_{\mathcal{D}_h^*, h}(x \mapsto \langle w, x \rangle) \geq \frac{1}{k} \geq \frac{1}{n+1}.$$

Proof. Proposition A.6 gives a nonnegative unit-sum zero representation of the signed vectors $h(q)q$. Apply Lemma A.7 to delete zero coefficients, duplicates, and then affine dependencies while preserving the vector sum and coefficient sum. The resulting positive support has size at most $n+1$. Proposition A.7 applied to that support gives the displayed uniform distribution and universal lower bound. The infeasibility condition is only the hypothesis of this proposition and is not used as a primitive assumption elsewhere. $\square$

A.6 Exactification by the Strict Accuracy Gap

Proposition A.9 (High-accuracy strict feasibility). *Under Assumptions 1, 2, and 3, for every fixed $h \in \mathcal{H}$, let $Q_h := \{q \in \mathcal{X} : h(q) = -s_0\}$. Then the strict system*

$$h(q)\langle w_h, q \rangle > 0 \quad (q \in Q_h)$$

has a solution $w_h \in \mathbb{R}^n$.

Proof. Fix h . Suppose for contradiction that the displayed strict system is infeasible. Proposition A.8 then supplies a distribution $\mathcal{D}_h^* \in \Delta(\mathcal{X})$ and an integer $1 \leq k \leq n+1$ such that, for every $w \in \mathbb{R}^n$,

$$\mathcal{L}_{\mathcal{D}_h^*, h}(x \mapsto \langle w, x \rangle) \geq \frac{1}{k} \geq \frac{1}{n+1}.$$

Apply Proposition A.4 to this same distribution and target. It yields a deterministic $v \in \mathbb{R}^n$ with

$$\mathcal{L}_{\mathcal{D}_h^*, h}(x \mapsto \langle v, x \rangle) \leq 2\varepsilon.$$

The lower bound is universal in w , so it applies to this same v . The identical target, distribution, score class, and tie-resolved loss therefore give the strict chain

$$\mathcal{L}_{\mathcal{D}_h^*, h}(\langle v, \cdot \rangle) \leq 2\varepsilon < \frac{1}{n+1} \leq \frac{1}{k} \leq \mathcal{L}_{\mathcal{D}_h^*, h}(\langle v, \cdot \rangle),$$

contradicting Assumption 2. No expectation is compared to a realized loss here: the deterministic vector was extracted before the contradiction. Thus the infeasibility hypothesis is false and a strict separator w_h exists. $\square$

Proposition A.10 (Exact identity representation for a fixed target). *Under Assumptions 1, 2, and 3, every fixed $h \in \mathcal{H}$ has a vector $w_h \in \mathbb{R}^n$ satisfying*

$$\forall x \in \mathcal{X}, \quad \text{sign}_{s_0}(\langle w_h, x \rangle) = h(x).$$

Proof. Fix $h \in \mathcal{H}$ and set $Q_h := \{q \in \mathcal{X} : h(q) = -s_0\}$. Proposition A.9 supplies a vector satisfying $h(q)\langle w_h, q \rangle > 0$ for every $q \in Q_h$. The reverse implication in Proposition A.5 extends its exact labels from each representative to its antipode and hence to all of $\mathcal{X}$. Strictness excludes zero scores on representatives, so the fixed tie rule is preserved without any ordinary-sign replacement. $\square$

A.7 Common Identity Map and Dimension Bounds

Proposition A.11 (Common identity-map exact representation). *Under Assumptions 1, 2, and 3, the single map $\varphi_{\text{id}}(x) = x$ satisfies*

$$\forall h \in \mathcal{H} \exists w_h \in \mathbb{R}^n \forall x \in \mathcal{X}, \quad \text{sign}_{s_0}(\langle w_h, \varphi_{\text{id}}(x) \rangle) = h(x).$$

Proof. If $\mathcal{H} = \emptyset$, the target quantifier is vacuous and the map remains fixed. Otherwise fix an arbitrary $h \in \mathcal{H}$ and apply Proposition A.10. Since $\varphi_{\text{id}}(x) = x$,

$$\langle w_h, \varphi_{\text{id}}(x) \rangle = \langle w_h, x \rangle$$

for every x , and applying the same tie-resolved sign map gives the desired identity. The map contains no target, distribution, initialization, or trajectory-dependent choice; only w_h depends on the fixed target. $\square$

Proposition A.12 (Probability-one point-mass representation). *Let $\mathcal{P}_{\text{id}} = \delta_{\varphi_{\text{id}}}$. Under Proposition A.11, for every $\mathcal{D} \in \Delta(\mathcal{X})$ and $h \in \mathcal{H}$,*

$$\Pr_{\varphi \sim \mathcal{P}_{\text{id}}} \left[\inf_{w \in \mathbb{R}^n} \Pr_{x \sim \mathcal{D}} [\text{sign}_{s_0}(\langle w, \varphi(x) \rangle) h(x) < 0] = 0 \right] = 1.$$

Proof. The point mass is fixed solely by the identity map. For arbitrary $(\mathcal{D}, h)$, Proposition A.11 supplies w_h and hence

$$\text{sign}_{s_0}(\langle w_h, \varphi_{\text{id}}(x) \rangle) h(x) = h(x)^2 = 1 \quad (x \in \mathcal{X}).$$

The strict error event is therefore empty at $w = w_h$, so its probability is zero. Nonnegativity of every risk gives

$$0 \leq \inf_w \Pr_{x \sim \mathcal{D}} [\text{sign}_{s_0}(\langle w, \varphi_{\text{id}}(x) \rangle) h(x) < 0] \leq 0.$$

The infimum is exactly zero, and the sole map in the support of the Dirac law belongs to the event with probability one. If $\mathcal{H} = \emptyset$, the universal target assertion is vacuous. $\square$

Lemma A.9 (Deterministic representation implies confident representation). *For every class $\mathcal{H}$, every deterministic-admissible feature dimension is confident-admissible by taking a point mass on the same map. Consequently,*

$$\text{dc}^{1/2}(\mathcal{H}) \leq \text{dc}(\mathcal{H}).$$

Proof. Let d be deterministic-admissible, with common map $\varphi : \mathcal{X} \rightarrow \mathbb{R}^d$ and target-specific vectors u_h such that

$$\text{sign}_{s_0}(\langle u_h, \varphi(x) \rangle) = h(x) \quad (x \in \mathcal{X}).$$

Choose δ_φ before $\mathcal{D}$ and h . For every subsequent pair, the inner risk is zero at $w = u_h$, and nonnegativity makes its infimum zero; the point mass assigns probability one to that event. Thus every deterministic-admissible dimension is confident-admissible, and taking the least dimensions gives the inequality. The argument is vacuous but valid when $\mathcal{H} = \emptyset$. $\square$

Proposition A.13 (Exact dimension and parameter-count chain). *Under Proposition A.11, Lemma A.9, and $n, m, T \geq 1$, $S = m(n+1)$,*

$$\text{dc}^{1/2}(\mathcal{H}) \leq \text{dc}(\mathcal{H}) \leq n \leq S \leq TS.$$

Proof. Lemma A.9 gives the first inequality, and Proposition A.11 makes the identity map an admissible deterministic map into $\mathbb{R}^n$, so $\text{dc}(\mathcal{H}) \leq n$. Since $m \geq 1$,

$$n \leq n+1 \leq m(n+1) = S.$$

Since $T \geq 1$ and $S > 0$,

$$TS - S = (T-1)S \geq 0,$$

so $S \leq TS$. Combining the inequalities proves the chain. At $m = 1$, $T = 1$, or $n = 1$, these are respectively weak equalities or the displayed specializations $S = n+1$, $TS = S$, and $S = 2m$; no excluded boundary case is used. The empty-class case is covered by vacuity. $\square$

A.8 Proof of the Main Theorem

Proof. Under the three assumptions, Proposition A.10 gives, for every $h \in \mathcal{H}$, a vector w_h with exact identity-coordinate labels. Proposition A.11 identifies those coordinates with the single map φ_{id} . The point-mass conclusion follows from Proposition A.12, and the complexity and parameter inequalities follow from Proposition A.13. If $\mathcal{H}$ is empty, all target quantifiers are vacuous and the same dimension candidate and arithmetic remain valid. These are exactly the assertions of Theorem 3.1. $\square$